2-DOF Helicopter Testbed User Manual

Operation, Safety and GUI Guide

Figure 1. Helicopter Testbed Graphical User Interface

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1. Introduction

The 2-DOF Helicopter Testbed is an educational and research platform designed to investigate the dynamics and control of a nonlinear MIMO system. The platform enables real-time PID tuning, data logging, and response visualization through a custom Windows GUI.

2. Safety Instructions

⚠ General Safety

- Operate the testbed only on a stable, level surface.
- Keep hands, loose clothing, hair, and tools away from rotating propellers.
- Keep away from the testbed whenever PID control is enabled; the helicopter may move suddenly.
- Ensure no objects are present within the moving path before every experiment.
- Never touch the moving mechanism while motors are running.
- Disconnect AC power before performing maintenance.
- If abnormal vibration, smoke, or uncontrolled motion occurs, disable PID immediately, switch OFF the testbed, and disconnect power.
- The propellers rotate at high speed and may cause serious injury.
- Keep hands away from the propellers at all times.
- Never touch the testbed while the motors are running.

PID Control Safety

When PID control is enabled, the helicopter can move suddenly without warning.

- Keep a safe distance from the testbed whenever PID control is active.
- Never place your hands inside the motion envelope while PID is enabled.
- Disable PID before making any adjustments to the mechanical structure.

Before every experiment:

Ensure no tools or objects are inside the movement area.

Verify that cables do not interfere with the moving parts.

Confirm that the helicopter can move freely without obstruction.

Emergency Procedure

If abnormal behavior occurs immediately:

- Disable PID control.
- Switch OFF the testbed.
- Disconnect the AC power supply.
- Inspect the system before restarting.

3. Testbed Overview

The testbed consists of a two-degree-of-freedom helicopter with independent Pitch and Yaw axes controlled by an STM32 microcontroller.

4. Testbed Setup

1. Place the testbed on a rigid, stable surface.
2. Ensure no object is within the moving range.
3. Check all electrical connections.
4. Connect the USB cable to the computer.
5. Verify the helicopter arm can move freely.

5. Start-Up Procedure

6. Connect the AC power supply.
7. Verify the power supply indicator LED is ON.
8. Turn ON the testbed.
9. Confirm the controller power/status LEDs are illuminated.
10. Turn OFF the testbed.
11. Launch the GUI and verify communication.

6. GUI Overview

The GUI provides communication, set-point control, PID tuning, real-time monitoring, response plotting, performance metrics, report export, and session management.

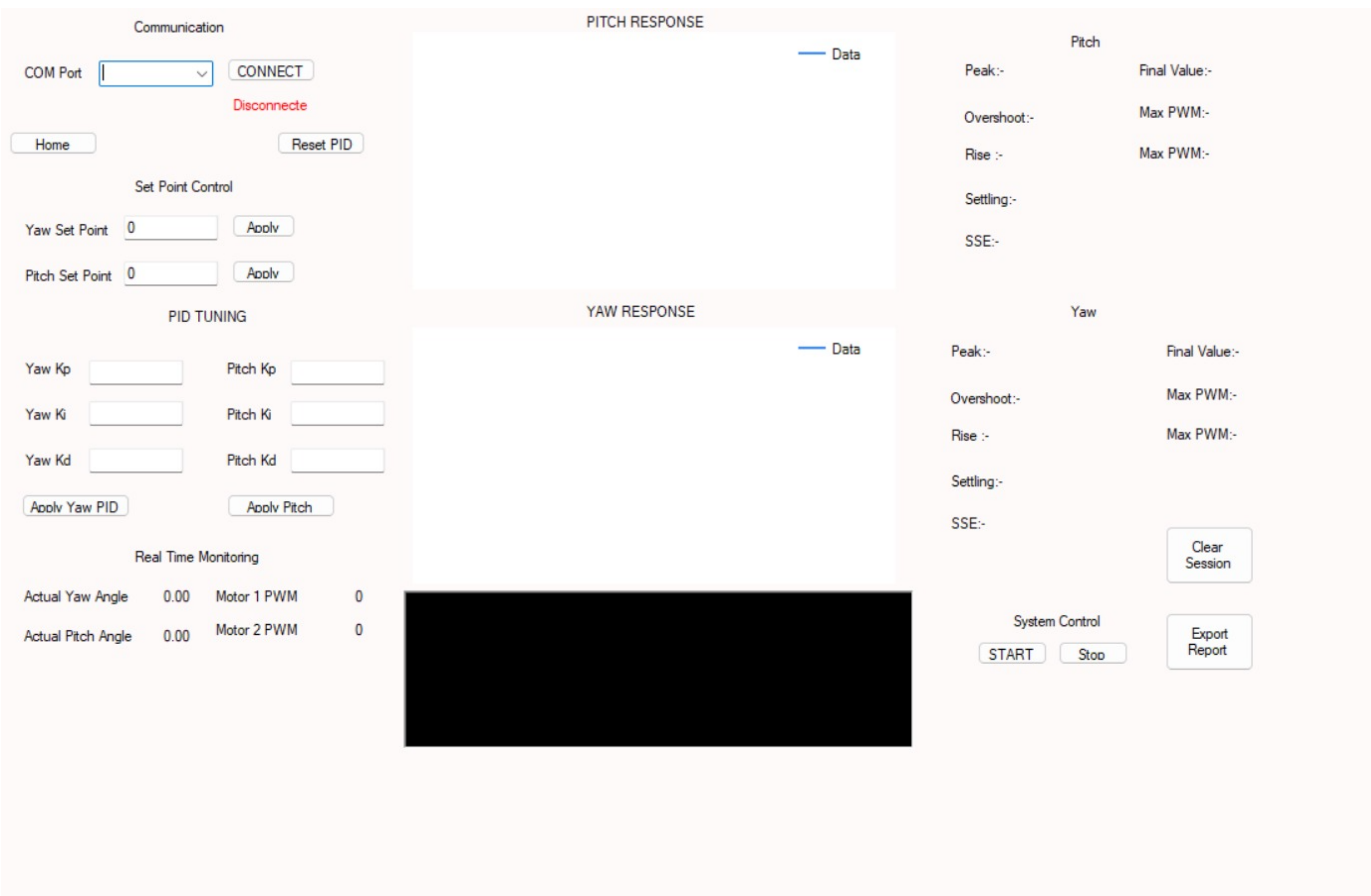
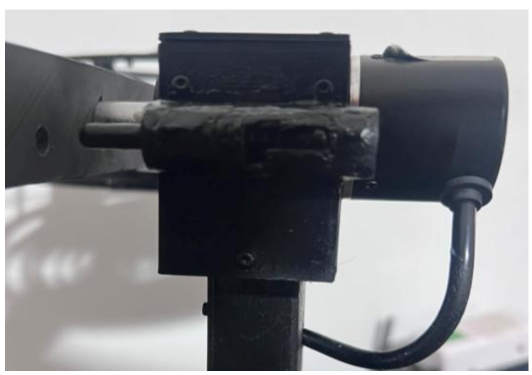


Figure 2: GUI Interface

7. Operating the Testbed

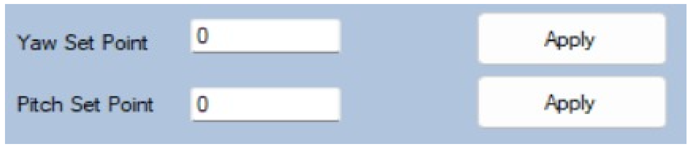
Connecting the GUI

1. Select the appropriate COM port.
2. Click **Connect**.
3. Verify that the connection status changes to **Connected**.
4. Confirm that live data begins updating.
5. Turn on the Testbed.
6. Then lock the pitch axis with the lock



7. Click **Home.**
8. Remove the lock

Changing Set Points



1. Enter the desired Pitch angle.

2. Click Apply beside the Pitch field.

3. Enter the desired Yaw angle.

4. Click Apply beside the Yaw field.

NOTE: Ensure the commanded values remain within the allowable operating limits. Only one setpoint value can be updated at a time

Enabling PID

1. Confirm the area around the testbed is clear.

2. Verify no person is within the moving area.

3. Click START.

4. The controller will begin tracking the selected set points.

Warning: Once PID control is enabled, the helicopter may move immediately.

Disabling PID

Click **Stop**.

Then Click **Stop**.

The controller will stop active position control and the motors will return to their safe state.

Changing PID Parameters

The GUI allows independent PID tuning for the Pitch and Yaw axes.

For each controller has:

- Kp
- Ki
- Kd

Click Apply Pitch PID or Apply Yaw PID only one value can be changed at a time

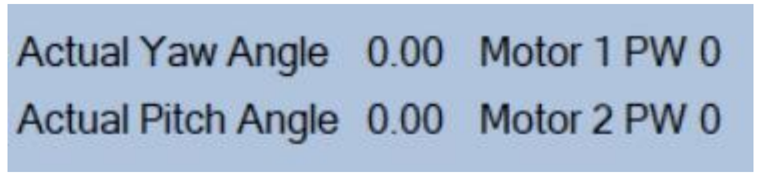
The updated controller gains are immediately transmitted to the embedded controller

Warning: Only one PID value can be updated at a time

Real-Time Monitoring

During operation, the GUI displays:

1. Actual Pitch Angle
2. Actual Yaw Angle
3. Motor 1 PWM
4. Motor 2 PWM



Exporting Report & CSV file

After completing an experiment:

1. Click Export Report.

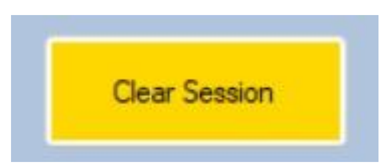
2. Choose a save location.

3. The experimental log will be saved for later analysis.

4. Do the same for export CSV

Clearing Session

Click Clear Session to clear graphs and statistics.



8. Shutdown Procedure

1. Press Stop.
2. Wait until motors stop completely.
3. Disconnect the GUI.
4. Close the application.
5. Turn OFF the testbed.
6. Disconnect AC power if no further testing is required.

9. Maintenance

- Inspect propellers and fasteners regularly.
- Keep the testbed clean and dust-free.
- Check connectors before use.